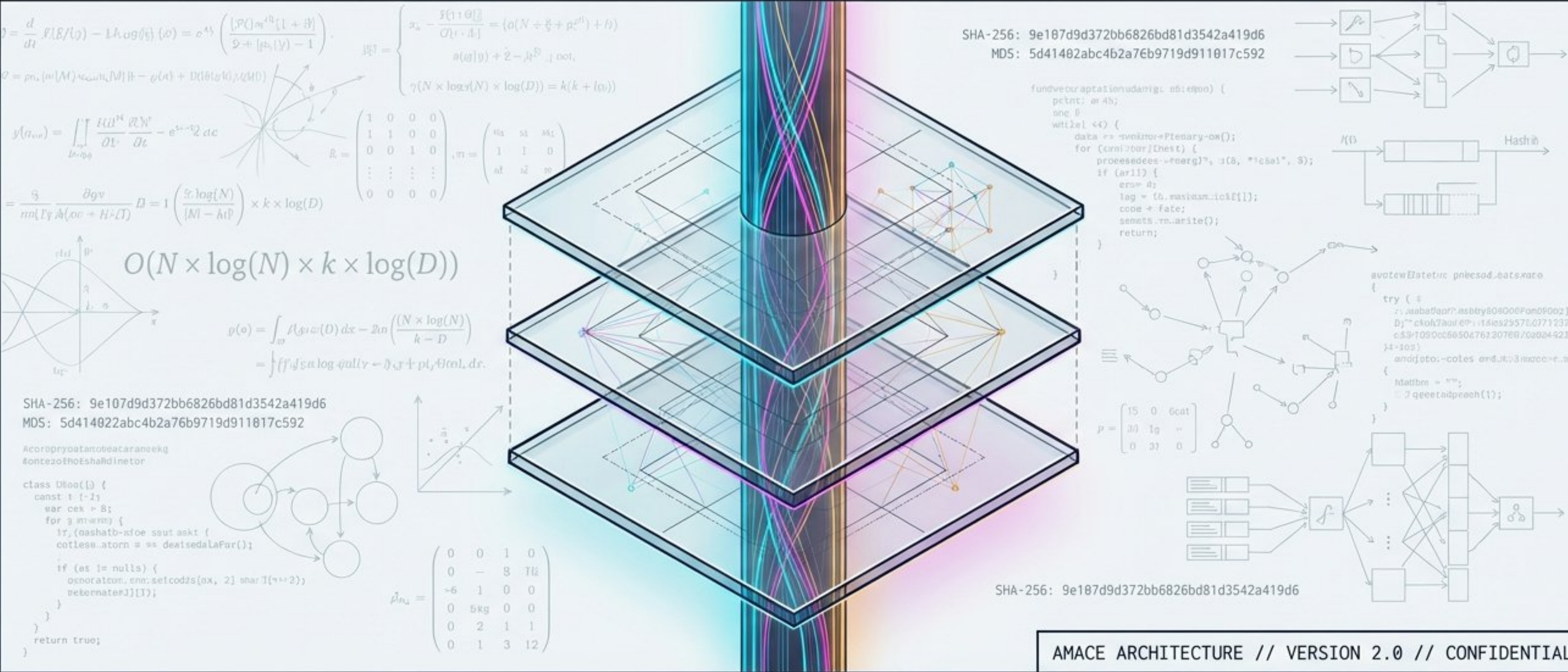


AUTONOMOUS MULTI-STRATUM ANOMALY CORRELATION ENGINE

DOMAIN-AGNOSTIC, ZERO-CONFIGURATION CAUSAL DISCOVERY ACROSS HETEROGENEOUS ABSTRACTION LAYERS



Every complex computing system operates across three fundamental abstraction strata

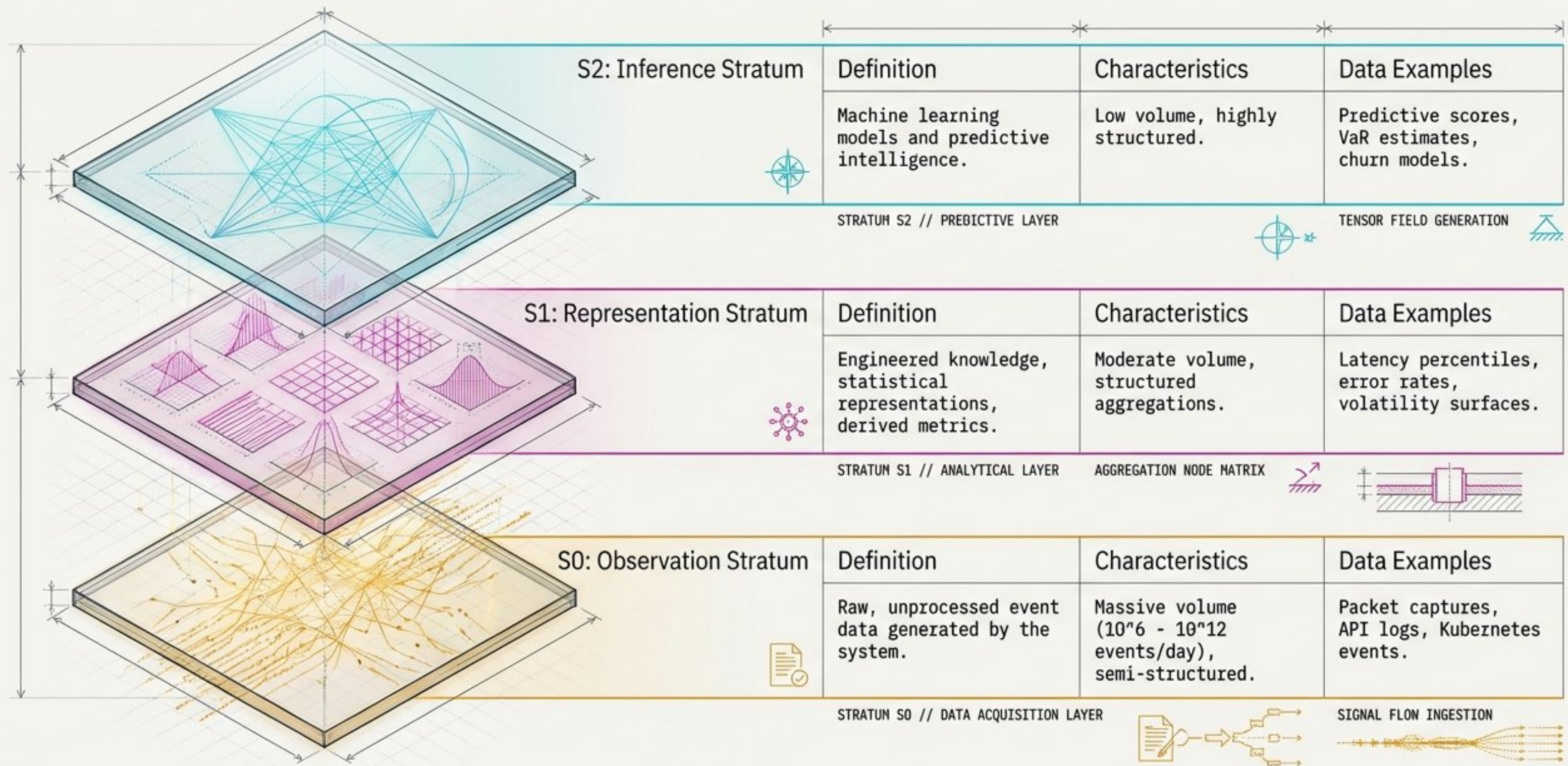
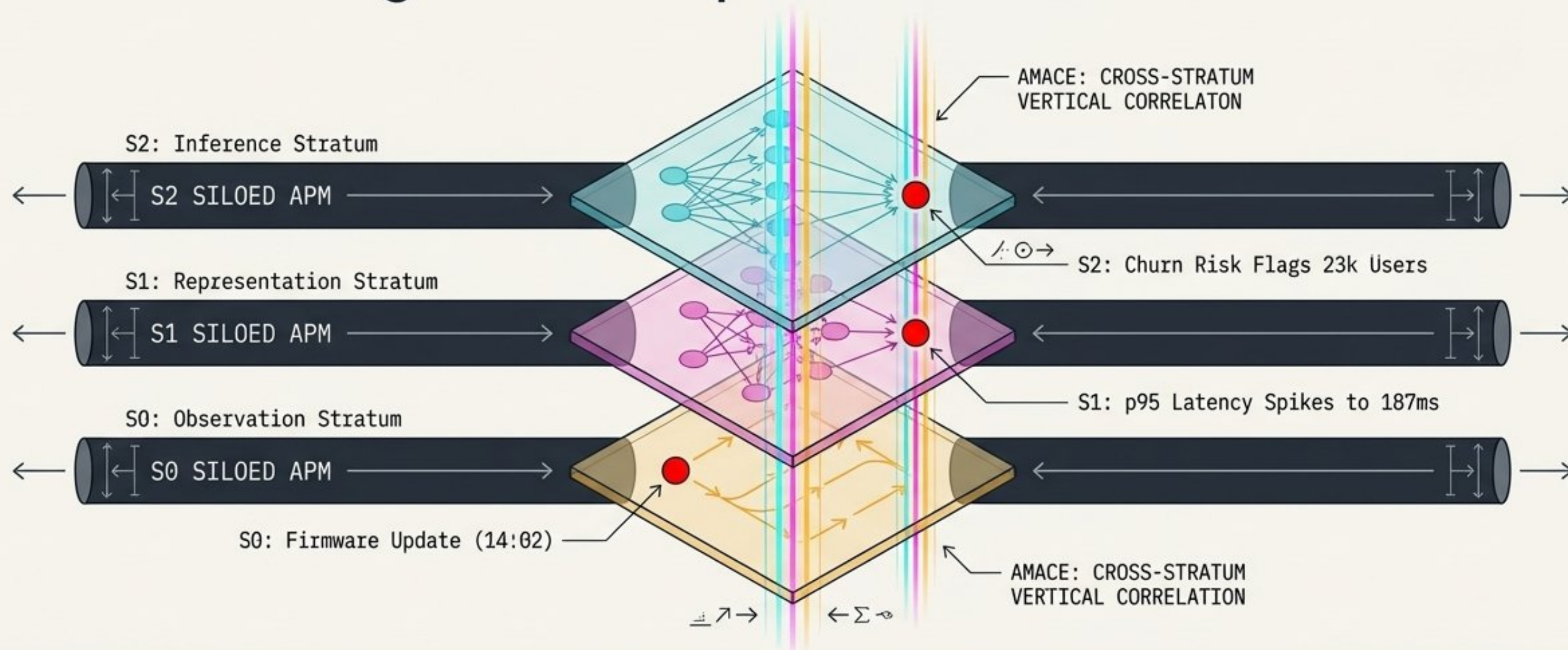


FIGURE 1.0: SYSTEM ABSTRACTION STRATA

The Cross-Stratum Blind Spot renders existing monitoring tools incapable of causal correlation

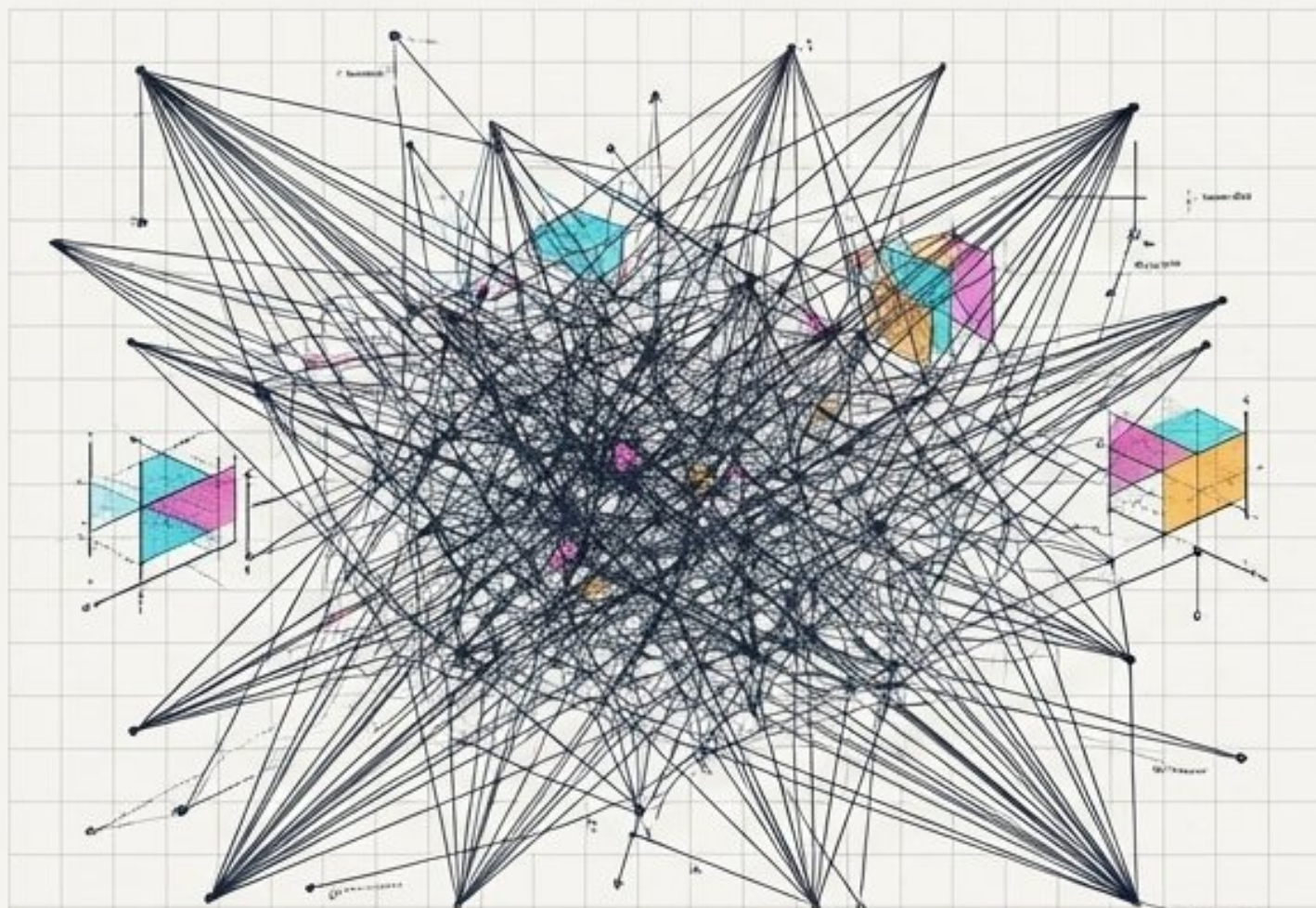


When an anomaly propagates across S0, S1, and S2, siloed tools cannot determine shared causal origins. Correlation requires vertical, cross-stratum analysis.

Cross-stratum correlation is computationally intractable without AMACE

Combinatorial Explosion

Given N anomalies across 3 strata, with D attributes and K cardinality.

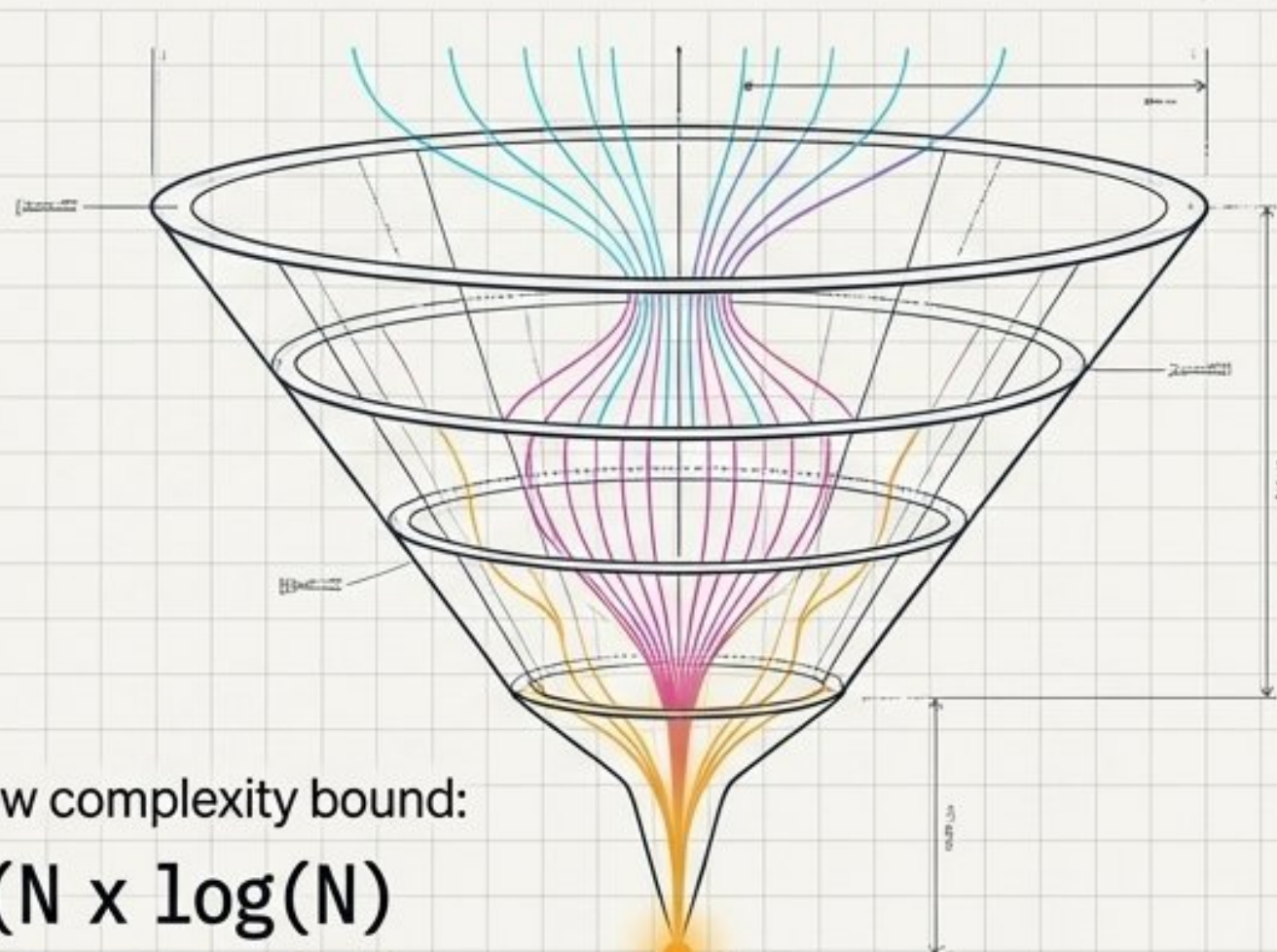


Total combinations: $O(N^2 \times K^D)$

Production Reality: $N=1000, D=8, K=50 \rightarrow >10^{18}$ combinations. Exhaustive search is impossible.

The AMACE Technical Improvement

Exponential reduction in required computational resources.



New complexity bound:

$$O(N \times \log(N) \times k \times \log(D))$$

Result: 10^{18} combinations mathematically reduced to roughly 40,000 evaluations.

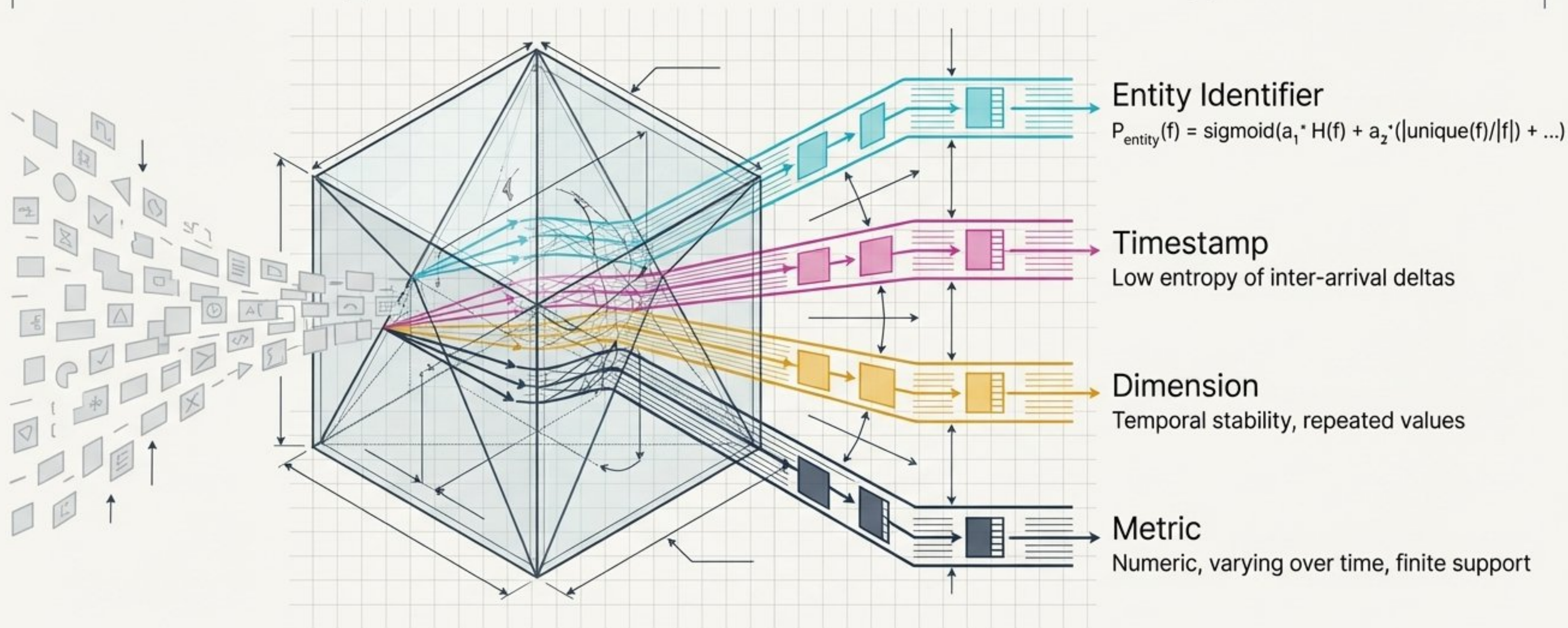
The paradigm shift from ad-hoc heuristics to formal algebraic structures



	Before (v1)	After (AMACE v2)
Terminology	Generic layers/signals	Terminology [Precise Strata and Cross-Stratum Causal Tensors]
Data Structure	Ad-hoc JSON fields	Data Structure [Formal Deviation Signature Tensor (7D tuple with algebraic operations)]
Dimensional Search	Conceptual MDL heuristics	Dimensional Search [Entropic Relevance Decomposition (Formal $O(k * \log(D))$ proof)]
Validation	Narrative guessing	Validation [Computational Ablation Protocol $\times$ (3 formal tests, statistical bounds)]
Setup	Generic Auto-discover	Setup [Stratum Schema Inference Engine (Entropy-based classification)]

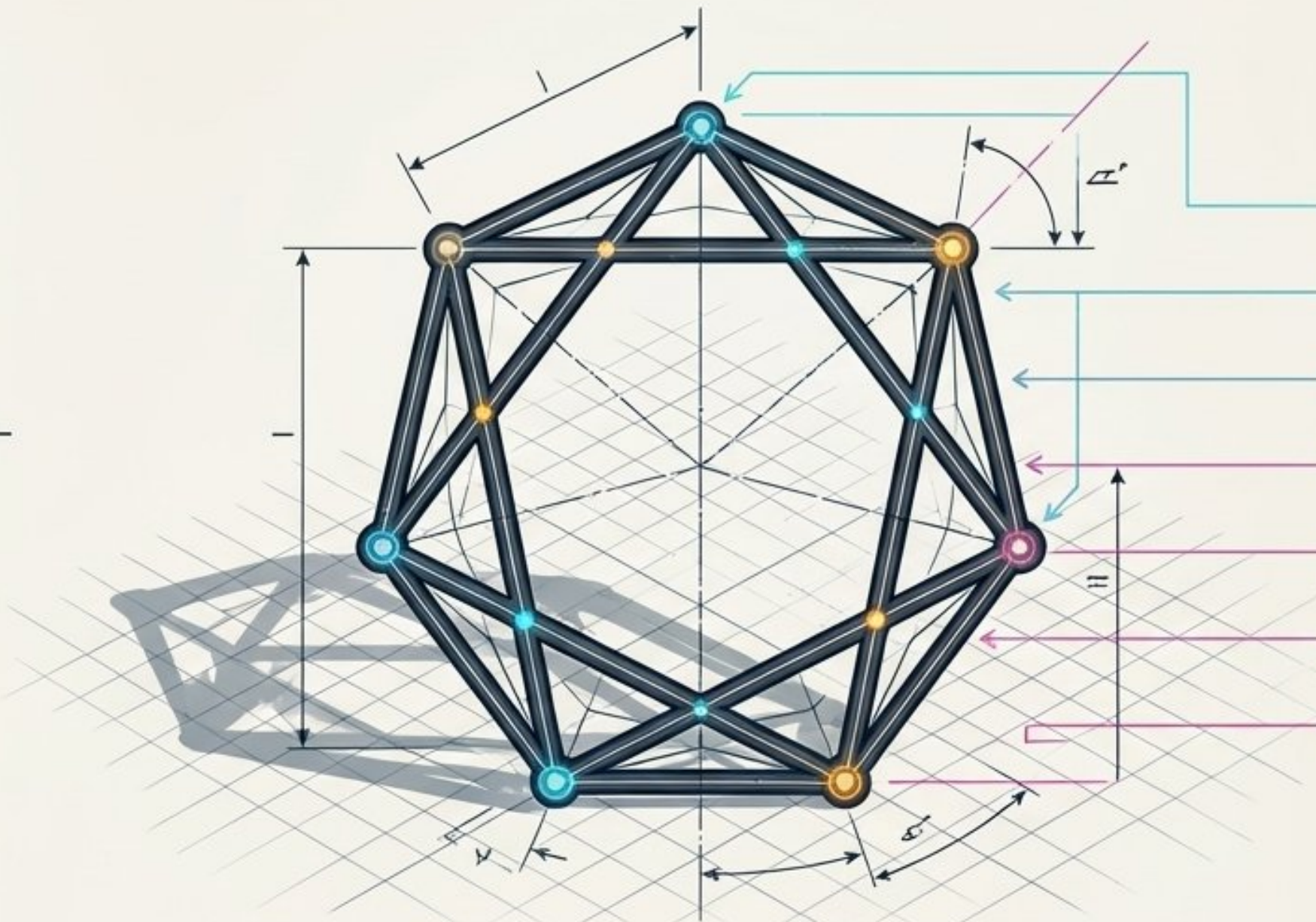
Module 1 classifies schema entirely through entropy and statistical fingerprinting

Zero configuration. No human input. No predefined rules.



Every detected anomaly is mapped to a universal Deviation Signature Tensor

SADD: Stratum-adaptive statistical detection (Modified Z-score with MAD for S0, PSI for S1, Prediction Interval Monitoring for S2).



S	(Stratum):	S0, S1, or S2 origin.
E	(Entity):	Resolved canonical identifier set.
T	(Time):	Onset, peak, decay, duration envelope.
D	(Dimension):	Affected sub-population locus.
M	(Magnitude):	Absolute, relative, and z-score severity.
C	(Confidence):	Statistical bounds $[u - z_a \cdot o, u + z_a \cdot o]$.
Φ	(Provenance):	SHA-256 hash for cryptographic audit trails.

$DST = (S, E, T, D, M, C, \Phi)$

IP ENFORCEABILITY:
Enables universal cross-stratum comparison impossible with prior art.

Entropic Relevance Decomposition isolates the exact dimensional slice of the anomaly

The Optimization

Maximizing $\text{ERD}(o) = \text{Surprise}(o) - \text{Cost}(o)$

Surprise: Information gained (bits) from localizing the anomaly.

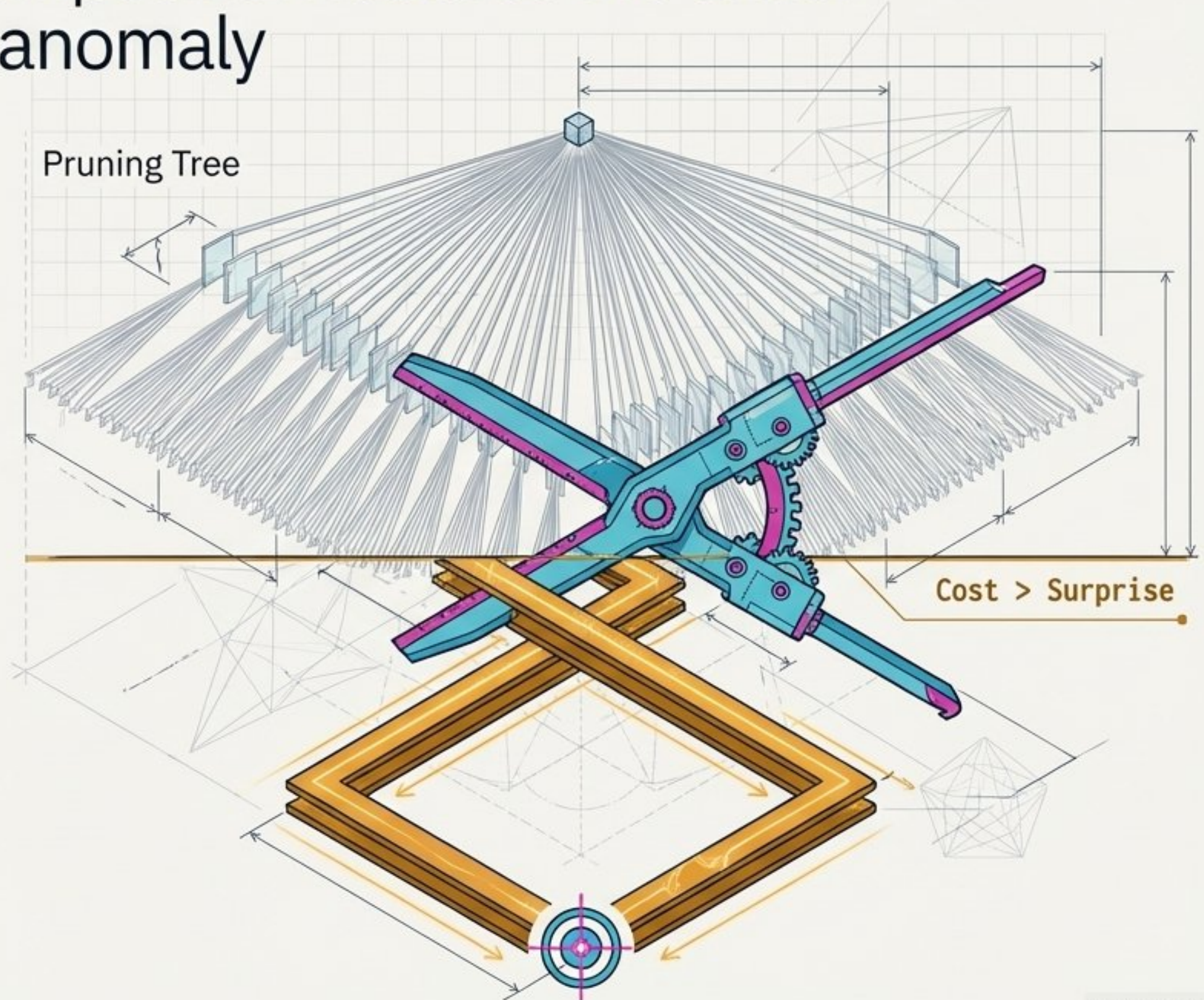
Cost: Information cost (bits) based on depth and cardinality.

Beam Search Mechanics

Instead of $O(100^8)$ exhaustive search, ERD uses width-limited beam search. Any candidate slice where **Cost > Surprise** is computationally pruned.

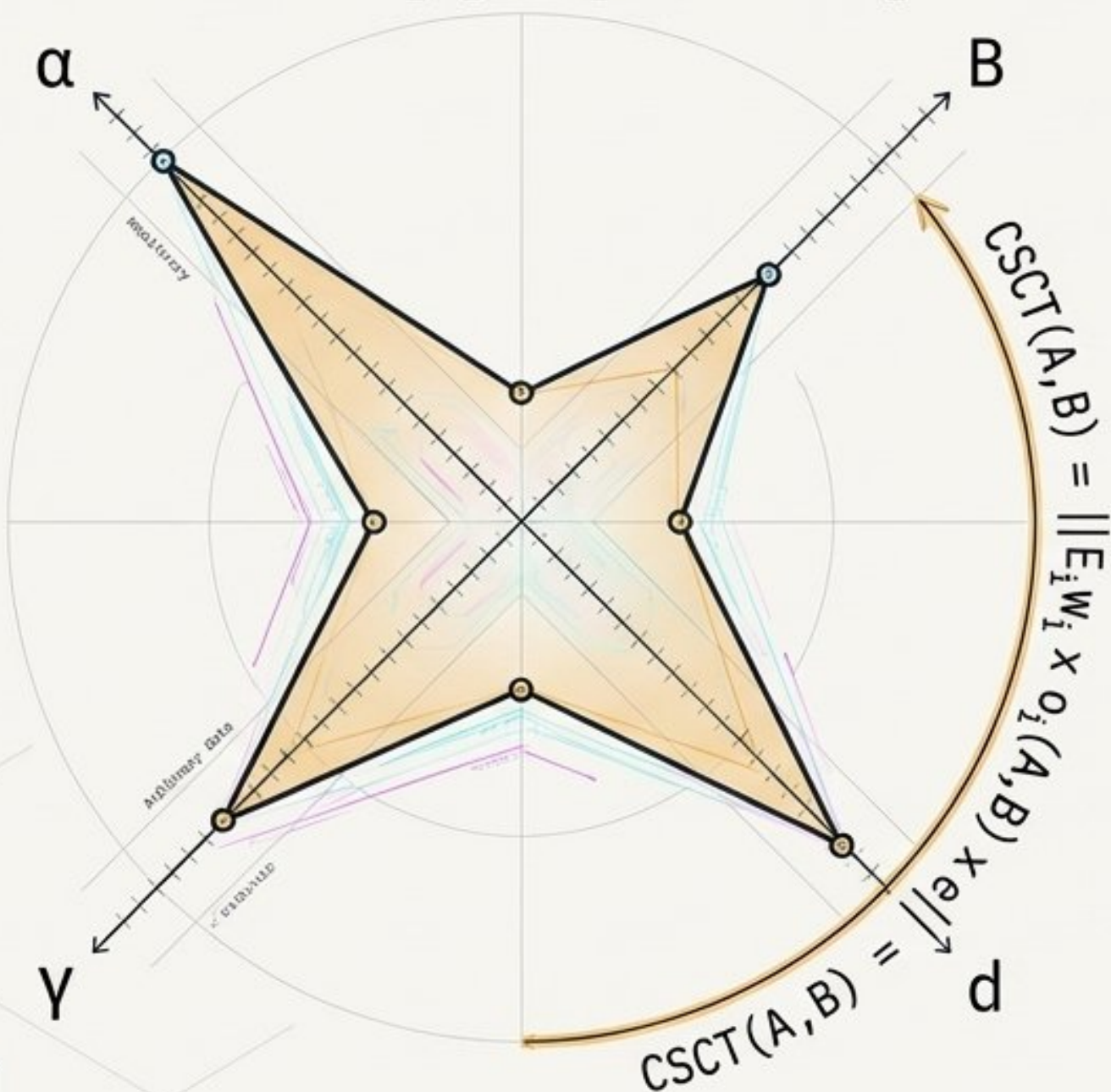
Result

Mathematically guaranteed to find the true dimensional locus in $O(w \times D \times K_{\max})$ time.



The Cross-Stratum Causal Tensor replaces ad-hoc correlation with orthogonal similarity scoring

$$\text{CSCT}(A, B) = ||E_i w_i \times o_i(A, B) \times e_i||$$



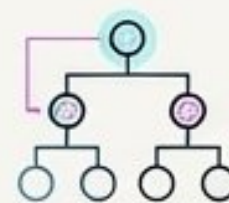
Axis 1: Entity Coincidence (α)

Surprise-adjusted Jaccard similarity. Overlap amplified when highly improbable under random chance.



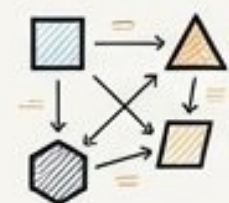
Axis 2: Temporal Precedence (B)

Gaussian kernel modeled on learned, dynamic propagation delays between strata, not fixed offsets.



Axis 3: Dimensional Coherence (γ)

Hierarchical matching (e.g., California and San Francisco score 0.5 as parent-child, not 0.0).

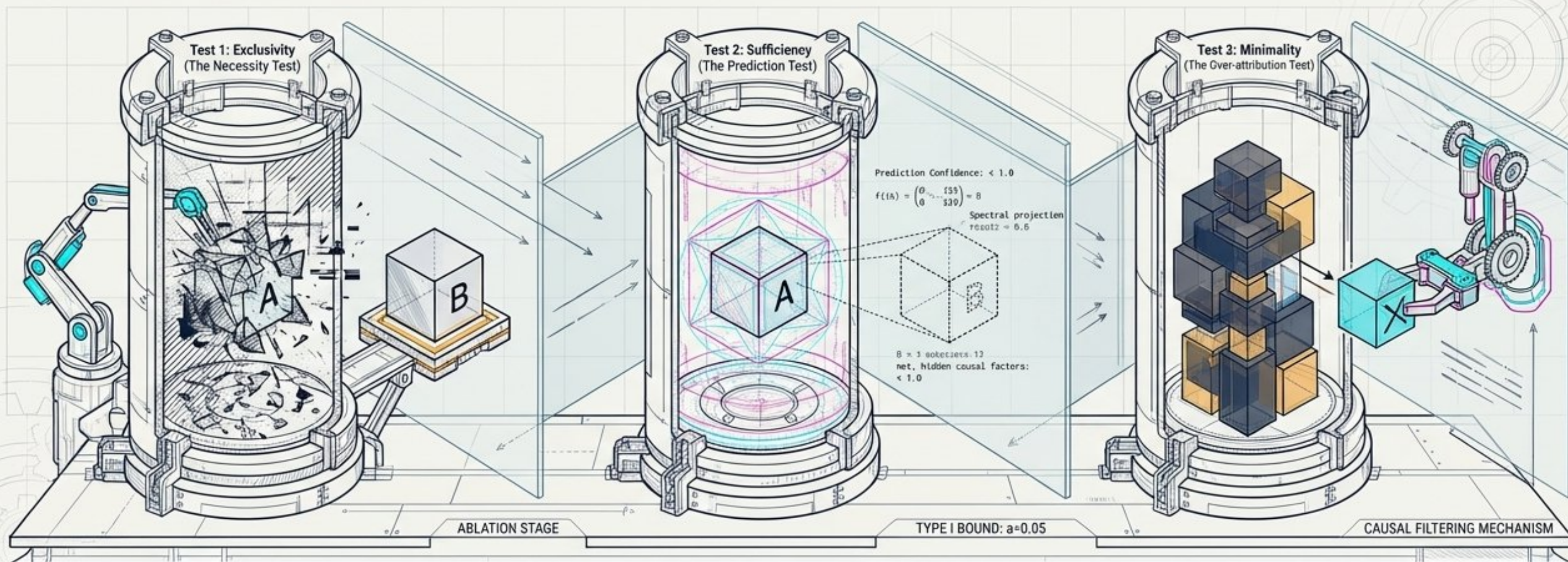


Axis 4: Semantic Affinity (d)

Contrastive-learning embeddings recognize metric relationships natively.

The Counterfactual Pivot: Systematically destroying data to mathematically prove causality

AMACE is the first system to perform automated counterfactual ablation on high-scoring correlations to prevent false causal claims. Bound at Type I error rate $\alpha = 0.05$.



Test 1: Exclusivity (The Necessity Test)

Delete Anomaly A. If Anomaly B persists unchanged, the correlation is coincidental. Pruned.

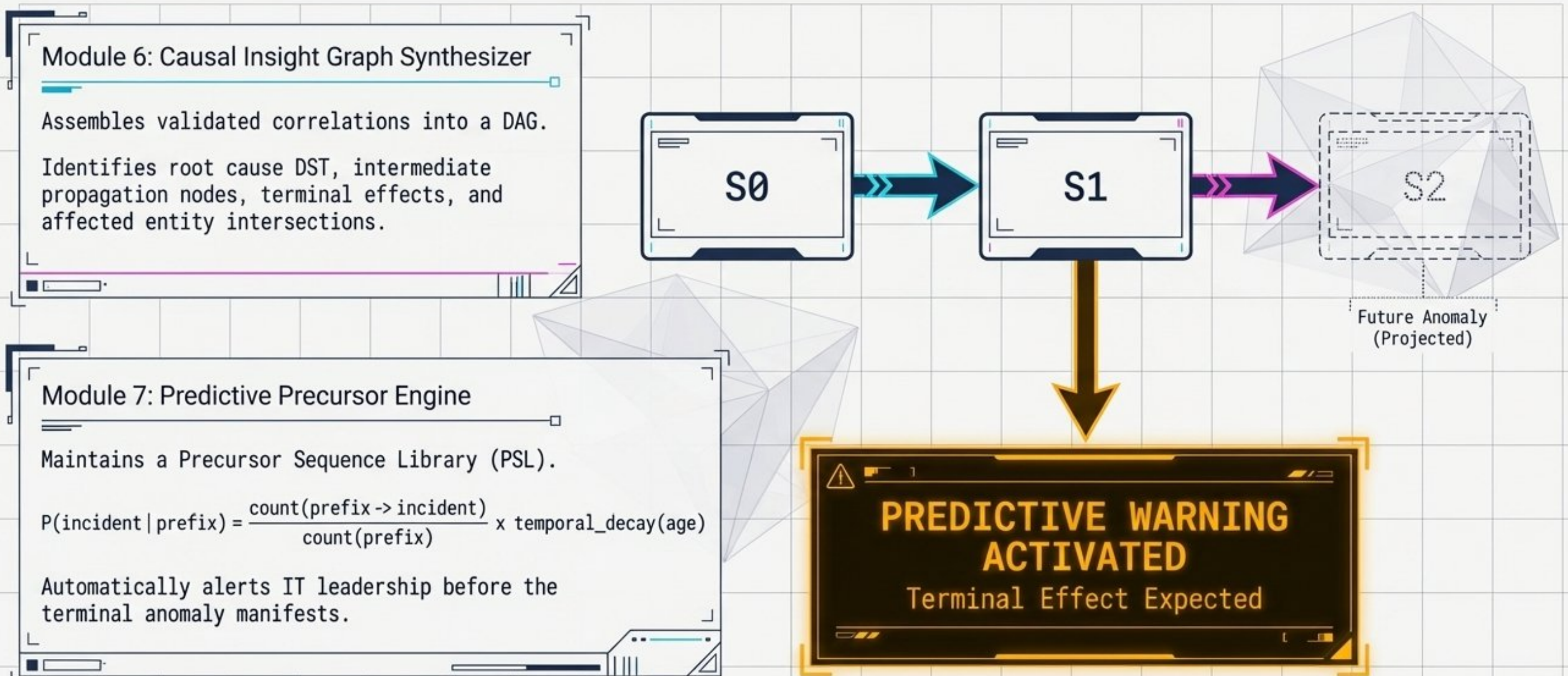
Test 2: Sufficiency (The Prediction Test)

Isolate Anomaly A. Can it mathematically predict the magnitude of Anomaly B? If not, hidden causal factors exist.

Test 3: Minimality (The Over-attribution Test)

Iteratively remove variables in complex chains to find the absolute minimum set required to explain the failure.

Causal Insight Graphs synthesize history to generate predictive warnings



Telecom Use Case: Resolving Cascading Service Degradation

The Symptoms (Disconnected)

S0:
RAN retransmissions
spike to 14.7%.

S0

S1:
Chicago p95 latency
jumps to 187ms.

S1

S2:
Churn model flags
23,000 subscribers.

S2

The AMACE Discovery

ERD identified: region=Chicago x equipment=edge_router x firmware=v3.2.1. (Pruned 99.94% of noise).
CAP confirmed: Firmware update was the sole necessary and sufficient cause.

The Impact

Time to root cause dropped
from 14 hours to 4.2 minutes.

~~14~~
HOURS



4.2
MINUTES

Time to root cause dropped
from 14 hours to 4.2 minutes.

Mobile Platform Use Case: Exposing Silent Revenue Loss

The Invisible Problem: A subtle recommendation API latency increase caused carousels to render late. No single metric breached outage thresholds.

Alert Threshold

S0: Minor API latency shift
(120ms → 380ms).

S1: Minor CTR drop to 4.1%.

S2: LTV prediction shifts downward
for Android mid-range cohort.

\$2.4M daily revenue loss

The Symptoms (Below Threshold)

S0: Minor API latency shift
(120ms → 380ms).

S1: Minor CTR drop to 4.1%.

S2: LTV prediction shifts downward
for Android mid-range cohort.

The AMACE Discovery

Entity coincidence axis ($a=0.97$) locked onto the exact user subset experiencing microscopic delays, linking the chain directly to a **\$2.4M daily revenue loss**.

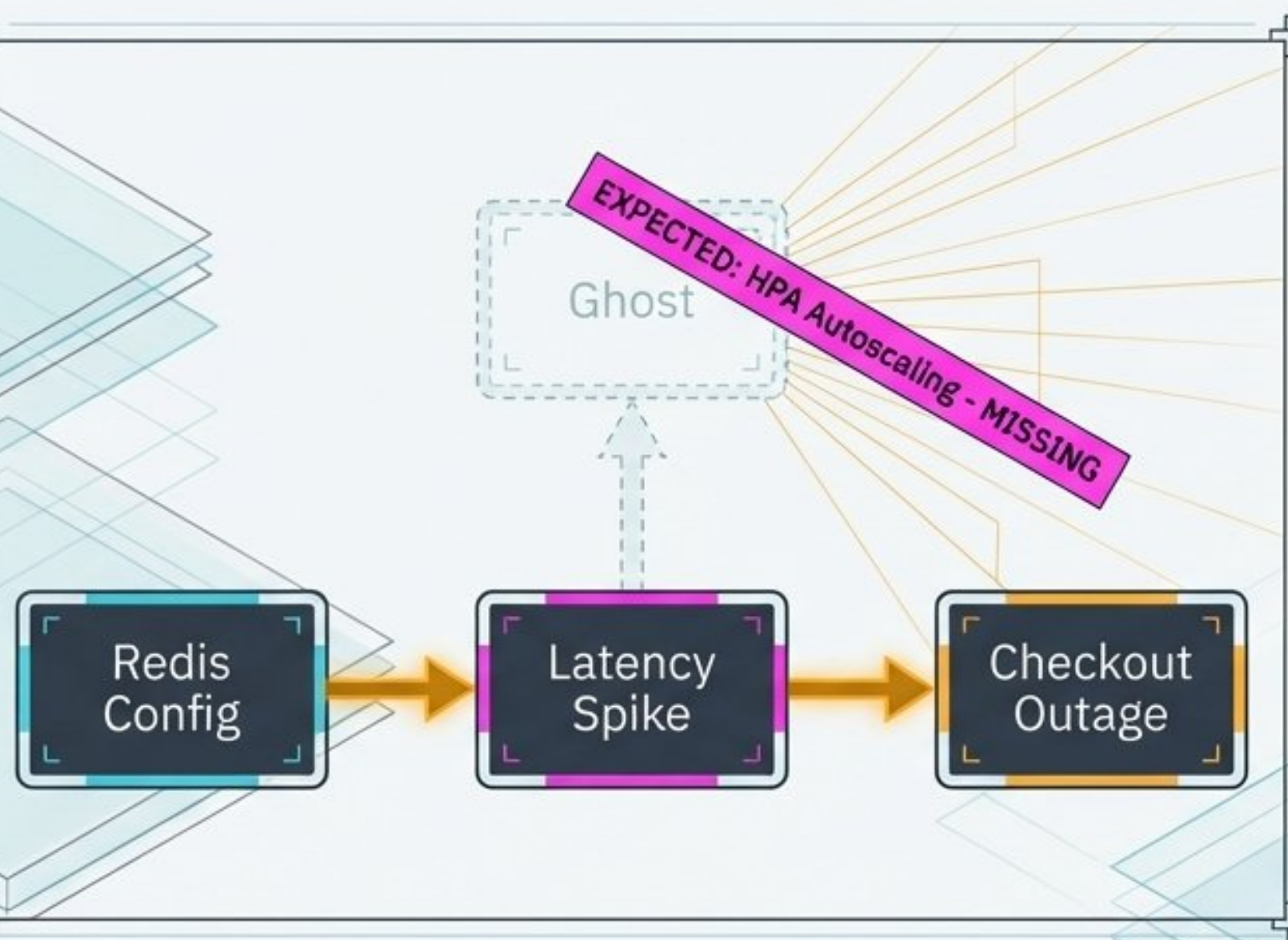
The Impact

Discovered a multi-million dollar leak in 12 minutes that standard APM tools were architecturally blind to.

Cloud Infrastructure Synthesis: The Anti-Correlation Revelation

The Incident

A Redis config change causes a memory leak, leading to microservice cascading failure and a checkout outage.



The Anti-Correlation Discovery

AMACE's predictive engine knew that a latency spike should have triggered horizontal pod autoscaling (HPA).

It didn't.

Dual-Insight Generation

AMACE did not just find the needle. It noticed a needle was missing from the haystack.



The Impact

Discovered both the primary root cause AND a latent, unknown misconfiguration in the autoscaler's health check endpoints in 8 minutes.



Financial Trading Use Case: Isolating Systemic Risk from Market Noise

The Chaos: Volatile market session triggers 47 independent risk alerts.

The Symptoms (Simultaneous)

S0: VIX spike & Reuters news wire (Fed unexpected statement).

S1: Cross-asset correlation matrix eigenvalue concentration increases.

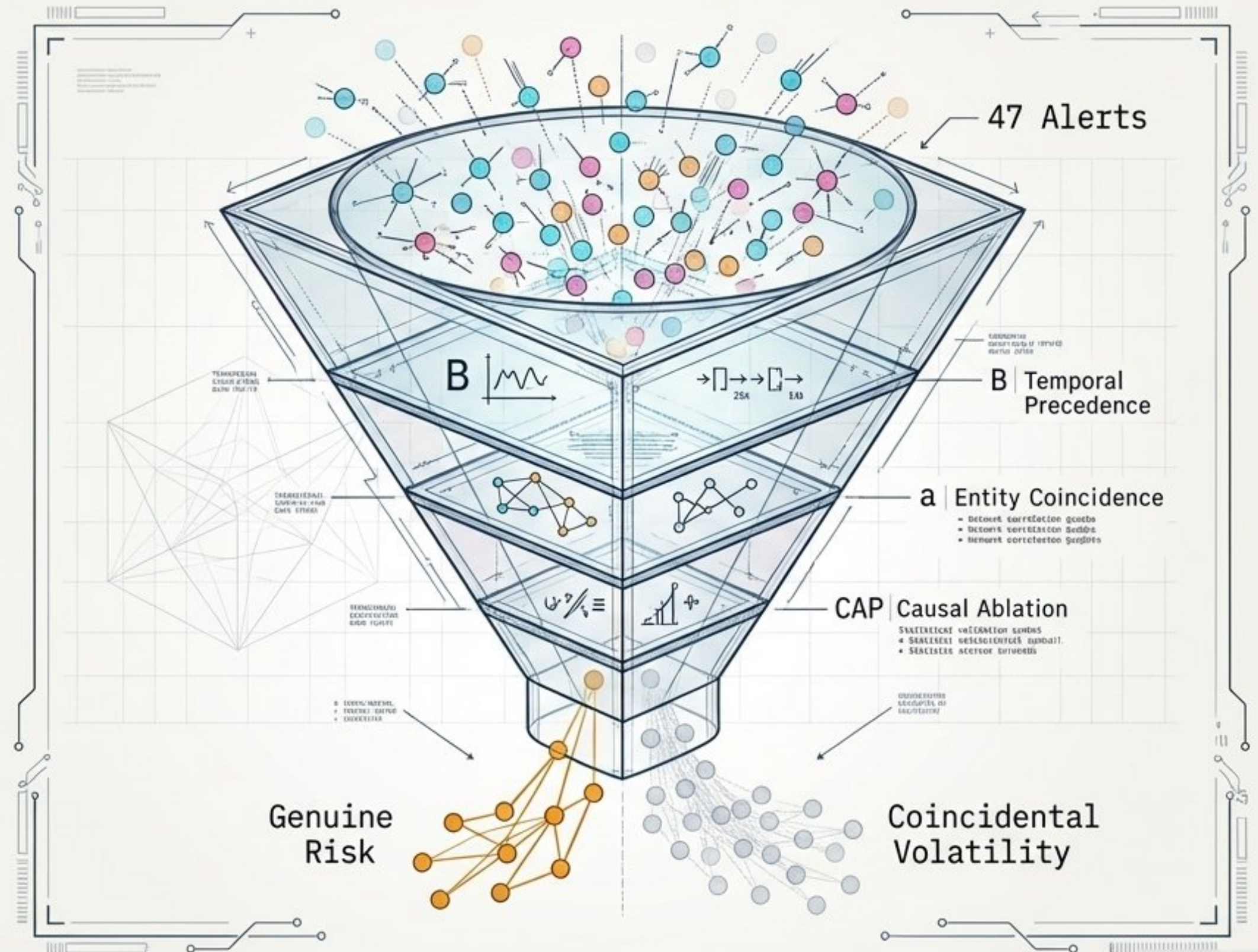
S2: 12 strategy signals reverse; VaR breaches 99.5th percentile.

The AMACE Discovery

Temporal Precedence (B) mathematically aligns microstructure delays:
news $\rightarrow$ 2.3s $\rightarrow$ yield shift $\rightarrow$ 14.0s $\rightarrow$ strategy reversals.

The Impact

CAP ablation validates exactly 7 genuinely correlated exposures. Prevented \$4.7M in unnecessary liquidation costs from 40 coincidental alerts.



Healthcare Clinical Use Case: Accelerating Adverse Event Detection

The Symptoms (Cross-Department)

- S0: Nursing medication overrides jump to 34%.
- S1: Post-surgical cardiac mean heart rate elevates.
- S2: Readmission model accuracy drops 8%.

The AMACE Discovery

ERD isolates the sub-population to **cardiac / post-surgical / 70-90kg**. A rounding error in the new protocol **under-dosed** this specific cohort.

Protocol
Deployment

AMACE Detection:
48 Hours

Manual Discovery:
11 Days

847 Patients
Protected

The Impact

Discovered the protocol defect in 48 hours instead of 11 days. Prevented 847 patients from receiving sub-therapeutic doses during the detection gap.

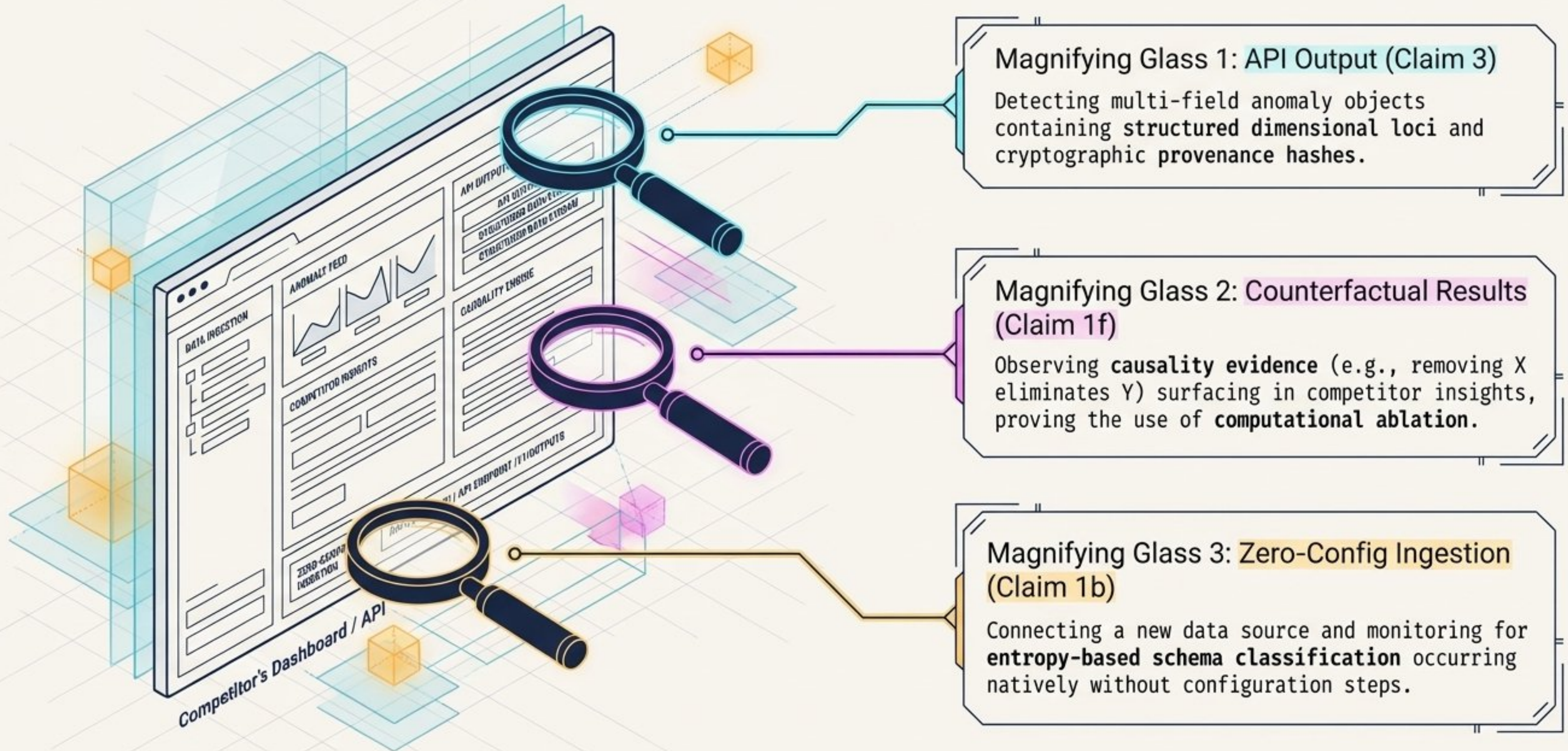
Detection Gap: 9 Days

Engineered from the ground up for bulletproof intellectual property defense



Statute	Typical Attack Vector	AMACE Technical Defense
Section 101 (Abstract Idea / Alice)	Merely correlating information.	Concrete data structures (7-element DST with provenance). Exponential reduction in computational complexity from $O(K^D)$ to $O(w * D * K_{max})$. An objective improvement to computer functionality.
Section 103 (Obviousness)	Predictable combination of ETL and correlation.	Unpredictable results (Anti-correlation, 99.97% search space pruning). Industry teaching-away: \$50B monitoring market is entirely built on isolated single-stratum tools.
Section 112 (Enablement)	Insufficient detail to implement.	Formal mathematical definitions, algorithm complexity bounds, and 5 extensively worked multi-domain examples providing constructive reduction to practice.

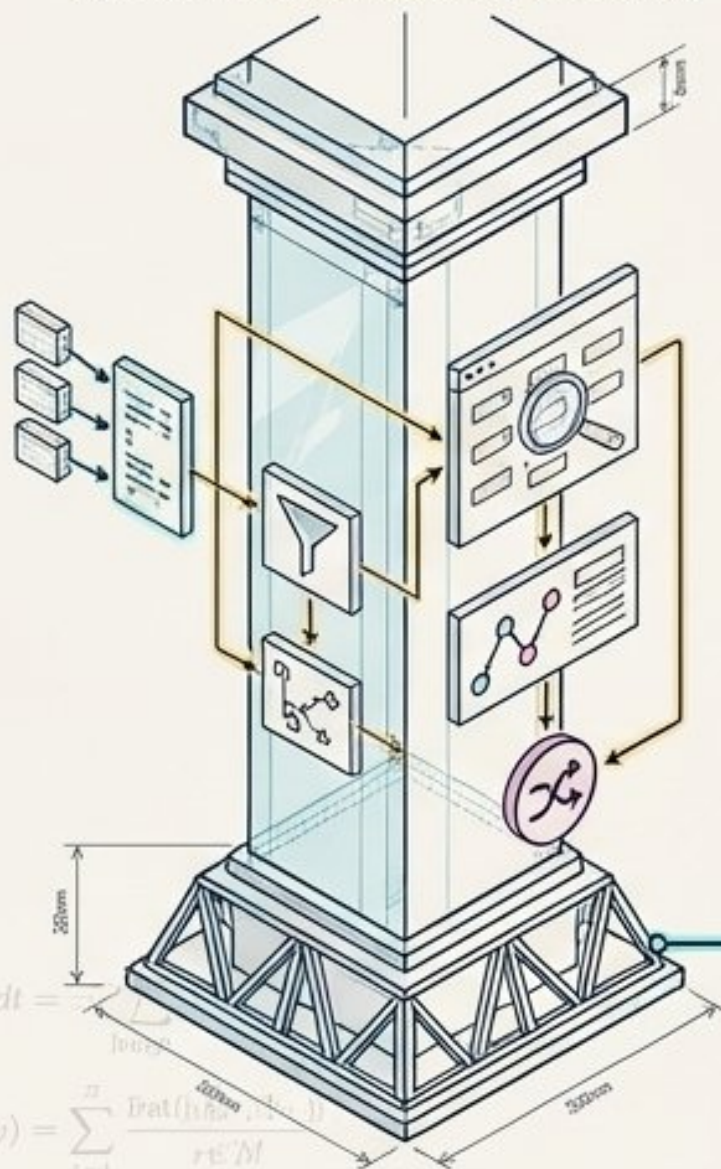
Externally observable outputs guarantee IP enforceability without source code



AMACE: The indisputable standard for cross-stratum causal discovery

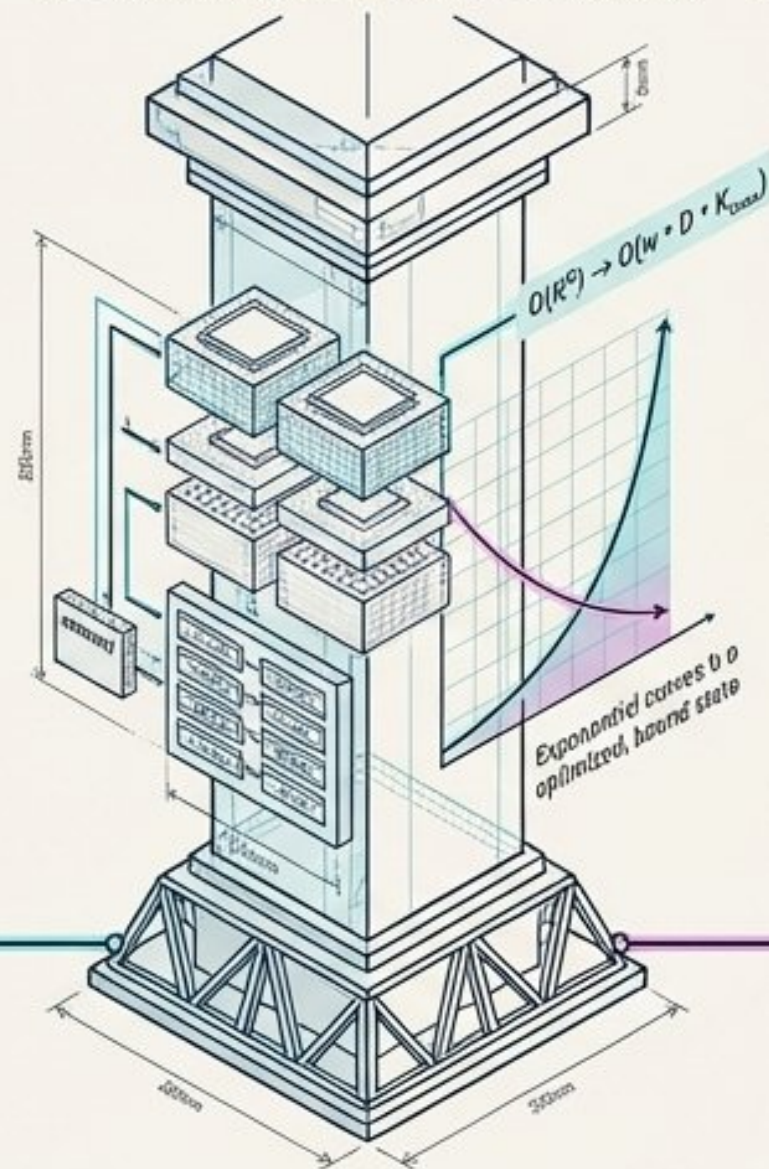
Pillar 1: Core Method (Claim 1)

The foundational pipeline from autonomous ingestion to counterfactual validation.



Pillar 2: System Construction (Claim 2)

The hardware/memory bounds executing the exponential search space reduction.



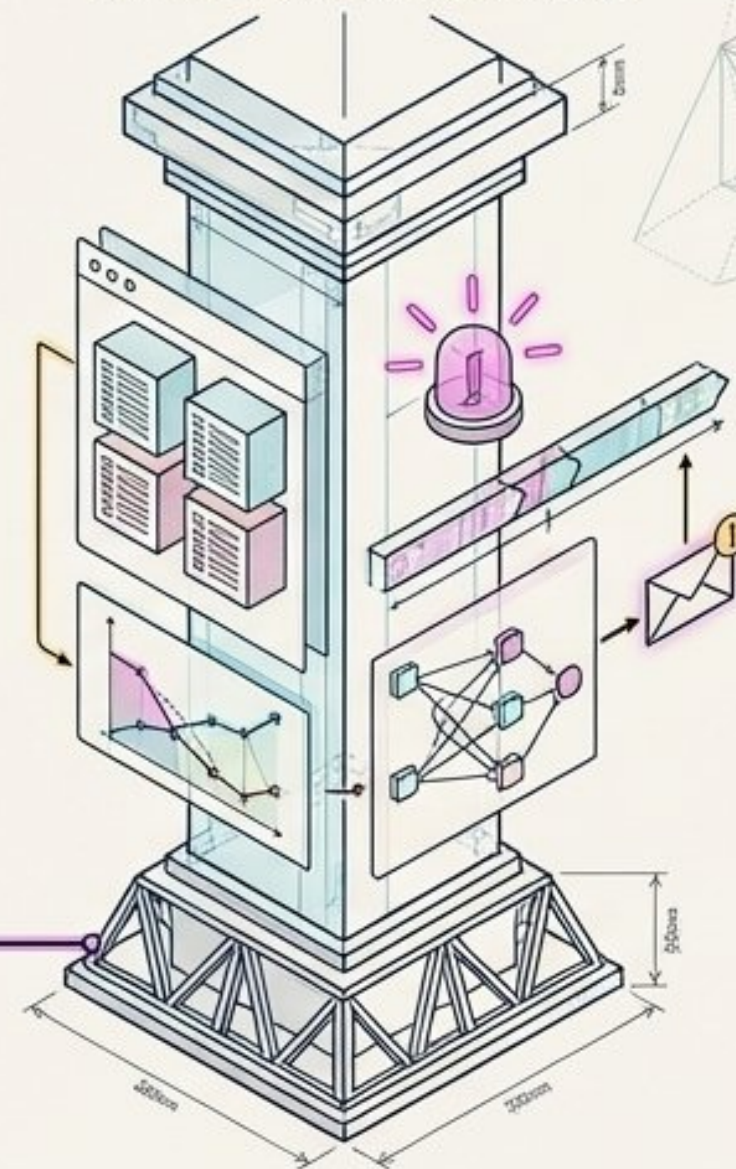
Pillar 3: Observable Output (Claim 3)

The structured causal insight DAG and UI components ensuring commercial trackability.



Pillar 4: Predictive Warnings (Claim 4)

The precursor sequence library and anti-correlation detection engine.



AMACE is not just a feature; it is a **mathematically proven, IP-protected paradigm shift** solving the most complex blind spot in modern computing.